the rebalancing strategy at time 2 (which are listed in equation (4.20)). That is, a short volatility position that is financed by bonds together with the buy-and-hold strategy is identical to the rebalanced strategy. Hence, *rebalancing is a short volatility strategy*.

In Figure 4.9, Panel D, the buy-and-hold strategy is the completely *passive* straight line. The rebalancing strategy is an *active* strategy that transfers payoffs from the extreme low and high stock realizations (S_{uu} and S_{dd}) to the middle stock realization ($S_{ud} = S_{dd}$). Rebalancing does this by selling when stock prices are high and buying when stock prices are low. Short volatility positions do exactly the same. A call option can be dynamically replicated by a long stock position and a short bond position. This buys equity when stock prices rise and sells equity when stock prices falls. A short call option does the opposite: a short call position is the same as selling when equity prices rise and buying when they fall. Likewise, a short put is also dynamically replicated by selling equity when prices rise and buying when prices fall. These are exactly the same actions as rebalancing.

3.2. INTERPRETATION

What is the market value of rebalancing? In this two-period binomial example, the action of rebalancing relative to the buy-and-hold strategy can be replicated by selling a call, selling a put, and investing in bonds. This has value:

$$\text{Short Call} + \text{Short Put} + \text{Long Bonds} = 0.0643 + 0.0428 - 0.1071 = 0.$$

That is, the action of rebalancing is assigned a zero market value. *The market does not value rebalancing*.

The optimal rebalancing strategy is a *partial equilibrium* strategy. Not everyone can rebalance. For every institution like Norway buying equities during the darkest periods of the financial crisis, there were institutions like CalPERS who couldn't wait to shed their risky equity allocations. CalPERS's losses in failing to rebalance represent Norway's gains from successful rebalancing. Put simply, for every buyer, there must be a seller. In equilibrium, it is impossible for everyone to simultaneously sell or buy.[28] Rebalancing is not valued by the market. In fact, consistent with this fact, the average investor who holds the market portfolio does not rebalance: *the market itself is buy and hold!*[29]

[28] Because of this, Sharpe (2010) advocates long-horizon investors to use "adaptive" asset allocation strategies that just drift up and down with the market instead of actively rebalancing. These are dominated, strictly, by rebalancing with i.i.d. returns as shown in section 2. See also Perold and Sharpe (1988).

[29] In the CAPM and in multifactor models, which we cover in chapter 7, the average investor holds the market portfolio. The average investor does not rebalance. Individual investors can rebalance only if some investors do not. Investors following portfolio insurance strategies, created by Hayne Leland